Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

6. Let C be the product of $A = \begin{bmatrix} 7 & 8 \\ 6 & 9 \end{bmatrix}$ with $B = \begin{bmatrix} 2 & 5 \\ 8 & 3 \end{bmatrix}$.

   (a) Use Strassen's method to compute C. Show all steps.

   (b) When using Strassen's method to multiply two n-by-n matrices, assume that the cost to add and subtract matrices is just n^2 and that the cost for the direct multiplication of two matrices is just n^3. Use these assumptions to derive a threshold on n for which Strassen's method becomes more advantageous than the direct multiplication.

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Answer:

6. (a)
   i. $A_{11} = [[7]]$, $A_{12} = [[8]]$, $A_{21} = [[6]]$, $A_{22} = [[9]]$,
   $B_{11} = [[2]]$, $B_{12} = [[5]]$, $B_{21} = [[8]]$, $B_{22} = [[3]]$.
   $S_1 = [[2]]$, $S_2 = [[15]]$, $S_3 = [[15]]$, $S_4 = [[6]]$, $S_5 = [[16]]$,
   $S_6 = [[5]]$, $S_7 = [[-1]]$, $S_8 = [[11]]$, $S_9 = [[1]]$, $S_{10} = [[7]]$.

   ii. $P_1 = [[14]]$, $P_2 = [[45]]$, $P_3 = [[30]]$, $P_4 = [[54]]$, $P_5 = [[80]]$, $P_6 = [[-11]]$, $P_7 = [[7]]$.

   iii. $C_{11} = [[78]]$, $C_{12} = [[59]]$, $C_{21} = [[84]]$, $C_{22} = [[57]]$,
   which corresponds to the product $\begin{bmatrix} 78 & 59 \\ 84 & 57 \end{bmatrix}$.

   (b) Under the assumptions, the cost of Strassen's method is

$$7\left(\frac{n}{2}\right)^3 + 10\left(\frac{n}{2}\right)^2 + 8\left(\frac{n}{2}\right)^2,$$

   which we compare against n^3, the cost of the straightforward multiplication:

$$n^3 = \frac{7}{8}n^3 + \frac{18}{4}n^2 \quad \Leftrightarrow \quad n^2\left(\frac{1}{8}n - \frac{9}{2}\right) = 0.$$

   Therefore, the threshold dimension is $n = 36$.
   Under the assumptions, Strassen's method becomes more advantageous for $n > 36$.